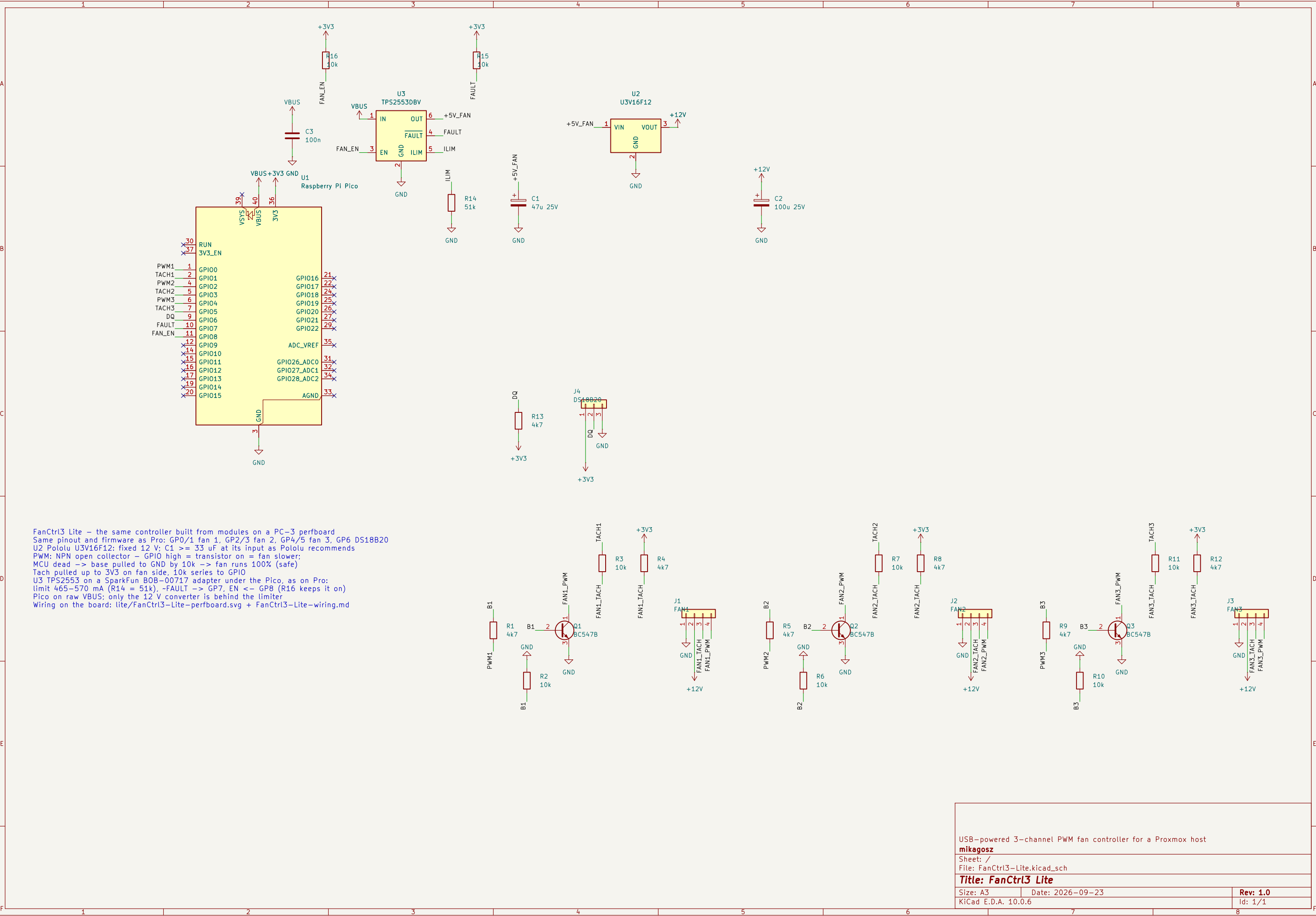




FanCtrl3 Lite – the same controller built from modules on a PC-3 perfboard
Same pinout and firmware as Pro: GP0/1 fan 1, GP2/3 fan 2, GP4/5 fan 3, GP6 DS18B20
U2 Pololu U3V16F12: fixed 12 V; C1 >= 33 uF at its input as Pololu recommends
PWM: NPN open collector – GPIO high = transistor on = fan slower;
MCU dead -> base pulled to GND by 10k -> fan runs 100% (safe)
Tach pulled up to 3V3 on fan side, 10k series to GPIO
U3 TPS2553 on a SparkFun BOB-00717 adapter under the Pico, as on Pro:
limit 465–570 mA (R14 = 51k), -FAULT -> GP7, EN <- GP8 (R16 keeps it on)
Pico on raw VBUS; only the 12 V converter is behind the limiter
Wiring on the board: lite/FanCtrl3-Lite-perfboard.svg + FanCtrl3-Lite-wiring.md